

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

08

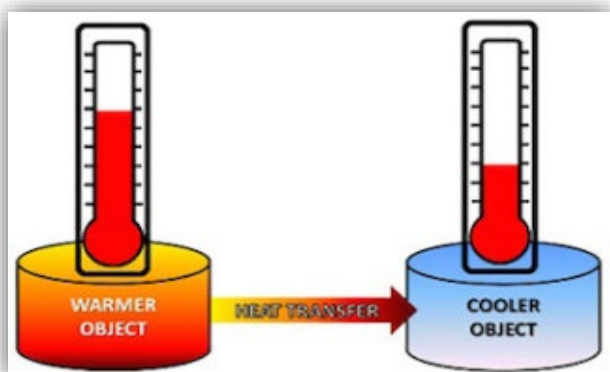
HEAT - 01



HEAT-01

Heat

- Heat is a form of energy that flows from a body at higher temperature to a body at lower temperature due to temperature difference.



SI Unit

- Joule (J)
- Another unit: Calorie:
 - 1 calorie = 4.186 joules

Nature of Heat

- Heat is energy in transit
- Not a property of a body Depends on mass, temperature change, and material

Temperature

- Temperature is the degree of hotness or coldness of a body.
- It determines the direction of heat flow (from higher to lower temperature).

SI Unit of Temperature

- Kelvin (K) → SI Unit

Other units:

- Celsius (°C)
- Fahrenheit (°F)

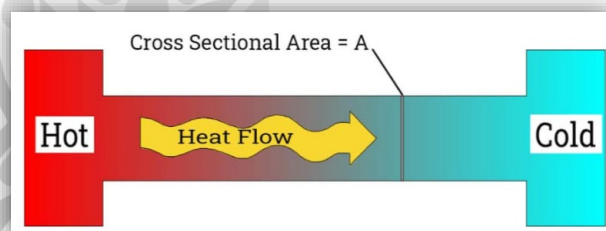
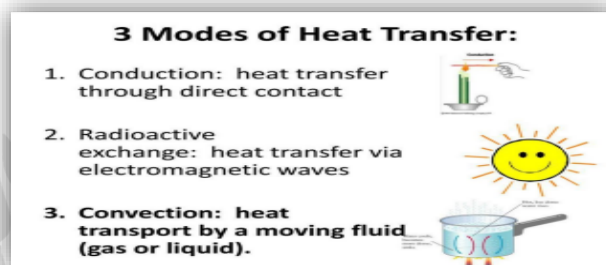
Difference Between Heat and Temperature

Heat	Temperature
Form of energy	Measure of hotness
Depending on mass	Independence of mass
Unit: Joule	Unit: Kelvin
Flows from one body to another	Indicates thermal state

Modes of Heat Transfer

I. Conduction

- Transfer of heat without movement of particles
- Occurs mainly in solids



Application of Thermal Conduction

- In winter, the iron chairs appear to be colder than the wooden chairs.
- Cooking utensils are made of aluminium and brass whereas their handles are made of wood.
- Ice is covered in gunny bags to prevent melting of ice.
- We feel warm in Woolen clothes and fur coat.
- Two thin blankets are warmer than a single blanket of double thickness.
- Birds often swell their feathers in winter.
- A new quilt is warmer than old one.

II. Radiation

- The process of the transfer of heat from one place to another place without heating the intervening medium is called radiation.
- When a body is heated and placed in vacuum, it loses heat even when there is no medium surrounding it. The heat cannot go out from the body by the process of conduction or convection since both of these processes require the presence



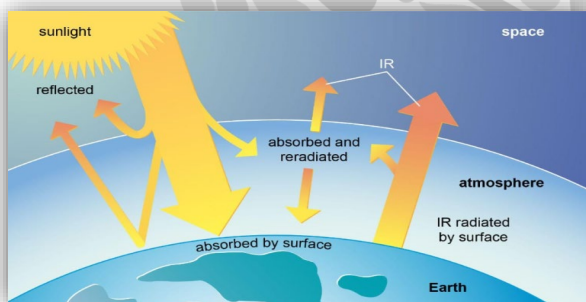


of a material medium between source and surrounding objects.

- When radiation passes through any medium then radiations slightly absorbed by medium according to its absorptive power so temperature of medium slightly increases.
- Radiation intensity is measured with a specific device named as **Bolometer**.
- Heat radiation are always obtained in infra-red region of electromagnetic wave spectrum so they are called Infra-red rays.

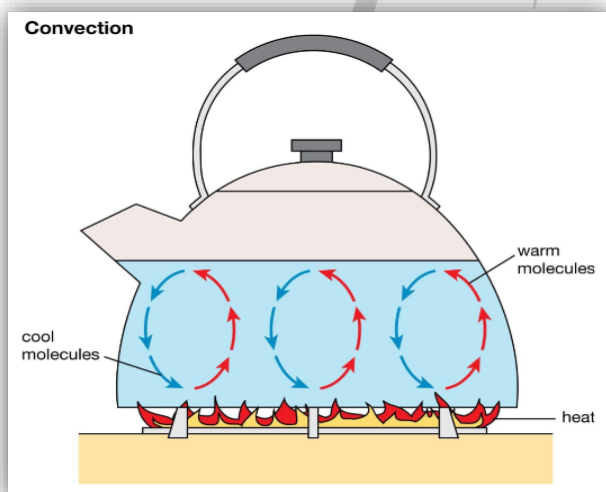
Application of thermal radiation:

- Heat from the Sun reaching Earth
- Solar cookers and solar panels
- Infrared heaters and thermal imaging
- Black surfaces absorb more heat (used in solar devices)
- Thermos flasks minimize heat loss via radiation



III. Convection

- Convection is the transfer of heat in liquids and gases due to the actual movement of particles.
- It occurs because of density differences created by temperature variation.



Application of Convection :

Land & Sea Breezes

- **Day:** Land heats faster → air rises → cool air from sea moves to land (sea breeze)
- **Night:** Land cools faster → air moves from land to sea (land breeze)

Trade Winds

- **Equator:** High temperature → low pressure (air rises)
- **Poles: Low temperature** → high pressure
- Air moves from poles to equator and gets deflected due to Earth's rotation
- Forms steady north-east trade winds

Monsoons

- Land heats more than ocean in summer
- Moist air from ocean moves to land
- Air rises, cools, condenses → causes rainfall

Ventilation

- Warm air rises and exits through ventilators/exhaust
- Fresh air enters from outside
- Based on forced convection

Specific Heat Capacity

- Amount of heat required to raise the temperature of 1 kg of a substance by 1°C.
- Where: $Q = MC\Delta T$
- Q = Heat
- M = Mass
- C = Specific heat capacity
- ΔT = Change in temperature

Latent Heat

- Heat absorbed or released without change in temperature during change of State.

Types of Latent Heat

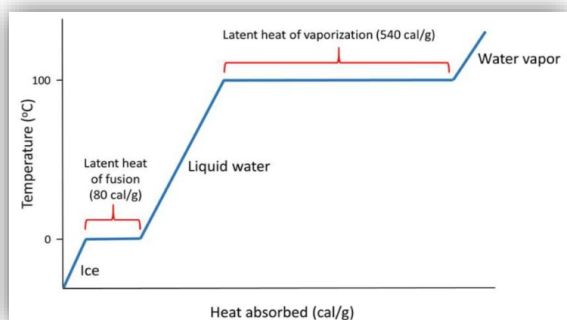
I. Latent heat of fusion

- Heat required to convert solid → liquid at constant temperature
- Example: Ice → Water

Key Points

- Temperature remains constant during melting
- Energy is used to break intermolecular forces
- Unit: J/kg





II. Latent heat of vaporization

- Heat required to convert liquid $\rightarrow$ gas at constant temperature
- Example: Water $\rightarrow$ Steam

Key Points

- Requires more heat than fusion
- Used to overcome strong intermolecular forces
- Responsible for cooling effect (evaporation)

Effects of Heat

- Increase in temperature
- Change in state latent Heat
- Expansion of matter
- Increase in molecular motion



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